Does the Distance Approach to Pairs Trading Survive in the Modern Era?

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The distance approach

- A pair trade holds offsetting positions in two assets whose prices share a long-run equilibrium, and profits when a temporary deviation closes.
- Gatev et al. (2006) trade pairs with the closest normalized price paths over a formation year, earning excess returns of up to 11% a year on all liquid US stocks from 1962 to 2002.
- Do and Faff (2010, 2012) found profitability declined to 2009, mainly because a growing share of pairs never converged, with the baseline rule largely unprofitable net of costs. They improved it by matching within industries and favoring pairs with many zero crossings.
- Krauss (2017) argued that ranking by the sum of squared distances selects low-variance spreads, whose trades earn little.

Research question

Question

1. Does the distance approach to pairs trading survive in the modern era, for US large-cap stocks at daily frequency from June 2014 to May 2024?
2. Does a composite score that adds zero crossings and spread volatility to the distance improve performance?

Finding

1. The distance approach does not survive.
2. No portfolio earns a mean return distinguishable from zero, with annualized means between -0.79% and 0.44%.
3. Neither the composite score nor the choice among Euclidean, Manhattan and Chebyshev distances makes a significant difference.

Data

Sector	Stocks	Pairs
Financial Services	24	276
Industrials	21	210
Healthcare	19	171
Consumer Cyclical	19	171
Utilities	18	153
Real Estate	17	136
Consumer Defensive	17	136
Technology	15	105
Communication Services	15	105
Basic Materials	12	66
Energy	12	66
Total	189	1,595

Stocks and within-sector pairs by sector.

- The universe is a list of 189 large-cap US stocks compiled in 2024.
- Daily adjusted closing prices run from June 2013 to May 2024.
- Pairs are formed within sectors, which gives 1,595 pairs.

Pair formation

The sample is divided into 20 consecutive trading periods E_k of 6 months, $k = 1, \ldots, 20$, from June 2014 to May 2024, each preceded by a 12-month formation period Φ_k.

$$\tilde{P}_{i,t} = \frac{P_{i,t}}{P_{i,\tau_W}}, \qquad d_{ij,t} = \tilde{P}_{i,t} - \tilde{P}_{j,t}$$

- $\tilde{P}_{i,t}$ and $d_{ij,t}$ are the normalized prices and the pair spread within a window $W \in \{\Phi_k, E_k\}$ beginning on day τ_W, with $P_{i,t}$ the adjusted closing price of stock i on day t.

$$D_{ij}^{(2)} = \Big(\sum_{t \in \Phi_k} d_{ij,t}^2 \Big)^{1/2}, \qquad D_{ij}^{(1)} = \sum_{t \in \Phi_k} |d_{ij,t}|, \qquad D_{ij}^{(\infty)} = \max_{t \in \Phi_k} |d_{ij,t}|$$

- These are the Euclidean, Manhattan and Chebyshev distances of a pair, eligible in period k when both stocks share a sector and a complete price record over Φ_k.
- The Euclidean distance ranks pairs as the sum of squared distances of Gatev et al. (2006).

Composite score

$$\frac{1}{T}\sum_{t\in\Phi_k} d_{ij,t}^2 = \frac{T-1}{T}\,\sigma_{ij}^2 + \bar{d}_{ij}^2$$

- T is the number of formation days, and σ_{ij} and $\bar{d}_{ij}$ are the standard deviation and the mean of the spread over Φ_k.
- A low distance therefore implies a low spread volatility, as argued in Krauss (2017).

$$R_{ij}^{(q)} = 0.6\,\phi\big(1/D_{ij}^{(q)}\big) + 0.2\,\phi\big(Z_{ij}\big) + 0.2\,\phi\big(\sigma_{ij}\big), \qquad q \in \{2, 1, \infty\}$$

- The composite score rewards volatility and zero crossings alongside a small distance.
- Z_{ij} is the number of sign changes of $d_{ij,t}$ over Φ_k, and ϕ scales a statistic linearly onto $[0, 1]$ over all eligible pairs.

Trading strategy

Prices are normalized again at the start of each trading period. The position $n_{ij,t} \in \{-1, 0, 1\}$ set at the close of day t follows the rule of Gatev et al. (2006),

$$n_{ij,t} = \begin{cases} -\operatorname{sgn}(d_{ij,t}) & \text{if } n_{ij,t-1} = 0 \text{ and } |d_{ij,t}| > 2\sigma_{ij}, \text{ and } t \text{ is not the last day of } E_k, \\ 0 & \text{if } n_{ij,t-1} \neq 0 \text{ and } d_{ij,t}\, n_{ij,t-1} \geq 0, \text{ or } t \text{ is the last day of } E_k, \\ n_{ij,t-1} & \text{otherwise.} \end{cases}$$

- A trade opened at the close of day o buys \$1 of the cheaper stock and sells \$1 of the other, then holds both until the spread crosses zero or the period ends.
- The portfolio return r_t averages its 20 pairs net of costs.
- The transaction cost is $c = 10$ basis points per dollar traded on each leg, so a round trip costs 40 basis points. The borrowing fee is $b = 25$ basis points a year on the short leg.

Hypotheses

Six portfolios

For each distance q and selection $m \in \{\text{distance}, \text{composite}\}$, the portfolio holds the 20 pairs with the lowest $D_{ij}^{(q)}$ or the highest $R_{ij}^{(q)}$.

Write $r_t^{(q,m)}$ for the daily return of the portfolio with distance q and selection m. We test

$$\begin{aligned}
&\text{H1}\colon\ \mathbb{E}\big[r_t^{(q,m)}\big] > 0,\\
&\text{H2}\colon\ \mathbb{E}\big[r_t^{(q,\text{composite})} - r_t^{(q,\text{distance})}\big] \neq 0,\\
&\text{H3}\colon\ \mathbb{E}\big[r_t^{(q,m)} - r_t^{(2,m)}\big] \neq 0.
\end{aligned}$$

- H1 asks whether each portfolio survives, H2 whether the composite score differs from the distance, and H3, for $q \in \{1, \infty\}$, whether the choice of distance matters.
- We test each with the t statistic for the mean daily return or difference, one-sided for H1 and two-sided for H2 and H3.

Net performance

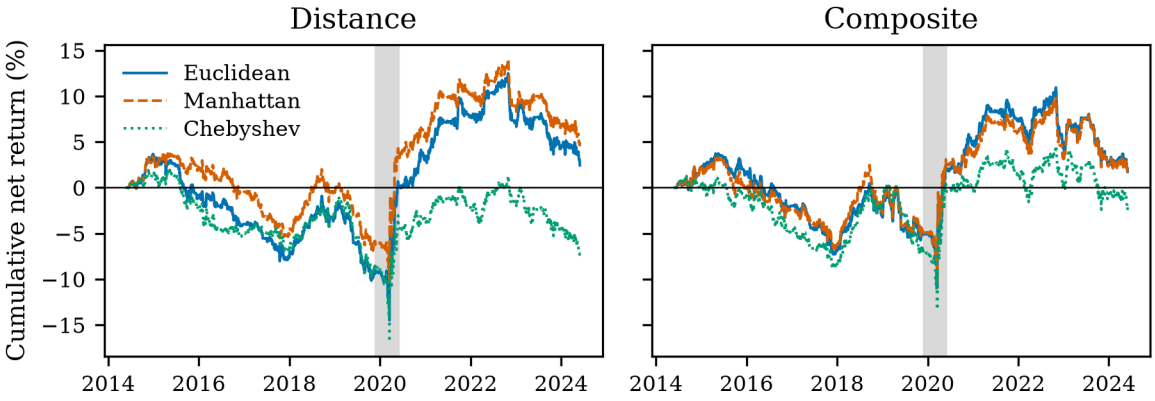
Distance	Selection	Return (%)	Volatility (%)	Sharpe	t	p
Euclidean	Distance	0.24	3.88	0.063	0.20	0.42
	Composite	0.17	3.98	0.043	0.14	0.45
Manhattan	Distance	0.44	3.91	0.11	0.36	0.36
	Composite	0.17	4.06	0.041	0.13	0.45
Chebyshev	Distance	−0.79	3.91	−0.20	−0.67	0.75
	Composite	−0.26	4.04	−0.065	−0.21	0.58

June 2014 to May 2024. Return and volatility are annualized, and p is the one-sided p value.

H1

- Distance-based pair formation does not survive in the modern era.
- With an annualized volatility of about 4%, no portfolio earns a return distinguishable from zero.

Cumulative net return



- The weakness is persistent rather than a product of the pandemic.
- The shaded COVID trading period from December 2019 to May 2020 first deepens the losses as spreads widen in March 2020, then recovers them sharply as the spreads close.

Returns before, during and after COVID

		Before COVID		COVID		After COVID		Excluding COVID	
Distance	Selection	Return	t	Return	t	Return	t	Return	t
Euclidean	Distance	−1.67	−1.41	18.9	1.07	0.58	0.33	−0.72	−0.71
	Composite	−0.92	−0.67	14.3	0.90	−0.056	−0.030	−0.56	−0.50
Manhattan	Distance	−1.10	−0.94	20.0	1.11	0.17	0.094	−0.57	−0.56
	Composite	−0.90	−0.64	15.2	0.93	−0.23	−0.12	−0.61	−0.54
Chebyshev	Distance	−1.59	−1.30	8.77	0.48	−0.85	−0.57	−1.28	−1.35
	Composite	−1.25	−0.96	13.1	0.74	−0.55	−0.32	−0.95	−0.91

Annualized net mean return (%) and t statistic from June 2014 to November 2019, in the COVID trading period from December 2019 to May 2020, from June 2020 to May 2024 and over all periods except the COVID period.

- All six portfolios lose money before COVID and over the 19 periods excluding COVID.
- The COVID period is the best for every portfolio, although its returns are not significant.
- The pandemic was not the cause of the poor performance but the one episode masking it.

Composite score and choice of distance

		Difference (%)	t	p
H2: composite minus distance selection				
Euclidean		−0.072	−0.12	0.90
Manhattan		−0.28	−0.40	0.69
Chebyshev		0.53	0.90	0.37
H3: Manhattan or Chebyshev minus Euclidean				
Manhattan	Distance	0.20	0.50	0.62
	Composite	−0.0059	−0.015	0.99
Chebyshev	Distance	−1.03	−1.60	0.11
	Composite	−0.43	−0.63	0.53

Differences in annualized net mean return, with two-sided p values.

- Neither refinement of the formation rule changes the conclusion.
- No difference is significant. The Chebyshev distance earns about 1% a year less than the Euclidean under the distance selection, with $p = 0.11$.

Formation statistics and trades

						Net profit per trade (%)		
Distance	Selection	σ (%)	Z	Trades/pair	Converged (%)	All	Converged	Forced
Euclidean	Distance	2.55	21.5	1.23	38.6	0.10	6.32	−3.81
	Composite	2.70	25.1	1.26	40.5	0.068	6.69	−4.44
Manhattan	Distance	2.59	22.3	1.23	39.6	0.18	6.43	−3.93
	Composite	2.78	25.8	1.26	39.5	0.066	6.92	−4.41
Chebyshev	Distance	2.58	18.5	1.16	36.1	−0.34	5.80	−3.81
	Composite	2.69	23.0	1.20	37.2	−0.11	6.30	−3.91

Z is the number of zero crossings. Trades per pair counts trades per selected pair and period. A trade is converged when the spread crosses zero and forced when it is closed at the end of the trading period.

- The composite score selects more volatile spreads with more zero crossings, yet opens barely more trades and closes a similar share by convergence.
- Only 36% to 41% of trades converge, and their gains of about 6% are offset by losses of about 4% on the rest.

Conclusion

- We formed pairs among 189 US large-cap stocks within sectors under three distances and ranked them by the distance alone or by a composite score.
- Over 20 periods of 6 months from June 2014 to May 2024, the rule of Gatev et al. (2006) does not survive, with annualized mean returns between -0.79% and 0.44%.
- Neither the composite score nor the choice of distance improves the result, and all six portfolios lose money without the COVID half-year.
- Rewarding spread volatility enlarges gains on converging and losses on non-converging trades alike, so it does not overcome the non-convergence that erodes the strategy.

References

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Code and replication data are available at *GitHub*.

Appendix A.1. Execution with a 1-day wait

Distance	Selection	Return (%)	Sharpe	t
Euclidean	Distance	0.073	0.019	0.060
	Composite	−0.062	−0.016	−0.051
Manhattan	Distance	0.36	0.097	0.30
	Composite	−0.13	−0.032	−0.10
Chebyshev	Distance	−1.02	−0.27	−0.91
	Composite	−0.50	−0.13	−0.42

Net performance with execution 1 day after the signal, June 2014 to May 2024.

- Executing a trade at the close that generates its signal may overstate returns through bid-ask bounce, which Gatev et al. (2006) address by waiting 1 day.
- This table repeats the analysis with every position change executed at the next close.
- Every portfolio earns less than in the net performance table, and no mean is distinguishable from zero.
- The composite score again makes no significant difference, with $|t| \leq 0.93$ for H2.